EXPERIMENT 03

Multiple Linear Regression

Python practical program using the California Housing dataset

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# Experiment 03: Multiple Linear Regression

# 1. Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# 2. Load dataset
data = fetch_california_housing()

df = pd.DataFrame(data.data, columns=data.feature_names)
df['MedHouseVal'] = data.target

# 3. Display first five records
print("First Five Records:")
print(df.head())

# 4. Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# 5. Separate input and output
X = df.drop('MedHouseVal', axis=1)
y = df['MedHouseVal']

# 6. Split dataset (80:20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 7. Create Multiple Linear Regression model
model = LinearRegression()

# 8. Train model
model.fit(X_train, y_train)

# 9. Predict values
y_pred = model.predict(X_test)

# 10. Calculate evaluation metrics
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

# 11. Display results
print("\nMean Squared Error:", mse)
print("Root Mean Squared Error:", rmse)
print("R2 Score:", r2)

# 12. Display intercept and coefficients
print("\nIntercept:", model.intercept_)
print("\nCoefficients:")
print(pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
}))

# 13. Actual vs Predicted Graph
plt.scatter(y_test, y_pred)
plt.xlabel("Actual House Value")
plt.ylabel("Predicted House Value")
plt.title("Actual vs Predicted Values")
plt.show()
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print("\nMultiple Linear Regression Completed Successfully!")
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